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Can we link emotional eating with the emotion regulation skills of adolescents?

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Objective: A recent cross-sectional study showed that maternal rejection is associated with emotional eating of obese youngsters seeking treatment, and that this relation is mediated by maladaptive emotion regulation (ER) of the youngsters. We wanted to build on this study and investigate the relation between parental rejection, maladaptive ER and emotional eating in a community sample using longitudinal data.

Design: Participants were 81 youngsters between the ages of 10 and 16 years. Participants completed questionnaires assessing maternal and paternal rejection, ER strategies and emotional eating, at two time moments ($M = 71$ days between time moments). Latent change models were used to estimate level and change of each variable.

Results: Results showed that the levels of maternal rejection, maladaptive ER and emotional eating were related. The indirect effect of the level of maternal rejection on the level of emotional eating through the level of maladaptive ER was marginally significant. On average, maternal rejection showed no change over time, whereas the other variables decreased. The changes in the variables were not related.

Conclusion: The findings highlight the importance of assessing the emotional bond between mother and youngster and the ER of youngsters with an emotional eating style.

Keywords: parental rejection; emotion regulation; emotional eating; adolescents; latent change model

Introduction

When confronted with severe distress, humans generally react with a ‘fight or flight’ response. This response implies a series of physiological reactions helping the person to deal with the threat, including the suppression of appetite (Charmandari, Tsigos, & Chrousos, 2005). However, research has found that experiencing less intense emotions, like being exposed to more chronic psychological stressors or to ordinary daily hassles, may lead to seeking out and consuming highly palatable food, such as food rich in sugar and/or fat (Macht, 2008; Macht & Simons, 2000; O’Connor, Jones, Conner, McMillan, & Ferguson, 2008; Oliver & Wardle, 1999). This eating behaviour has been

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defined as ‘emotional eating’ (van Strien & Oosterveld, 2008; p. 72) and is explained as a coping mechanism during moments of stress, when people try to ‘eat their troubles away’ with palatable food (Kaplan & Kaplan, 1957). Research has shown that eating palatable food can increase positive mood and reduce stress through sensory pleasure (Gibson, 2006), confirming the theory of Kaplan and Kaplan (1957). Unfortunately this increase in positive mood is only temporary and is mostly followed by negative feelings like shame (Gibson, 2006). Moreover, emotional eating may evolve into an eating disorder, such as binge eating disorder (BED), as the longitudinal survey study of Stice, Presnell, and Spangler (2002) showed that emotional eating is an important antecedent of binge eating. This disorder is characterised by eating large amounts of food in short periods of time while loss of control overeating is experienced (American Psychiatric Association, 2013).

Furthermore, as emotional eating can occur in the absence of hunger or can foster eating beyond the saturation point, it can lead to an increase in the calorie intake and in turn may lead to become overweight over time. However, research is inconsistent about the relation between emotional eating and body mass index (BMI). Regarding self-reported emotional eating, researchers have found positive association with BMI (e.g. Bryant, King, & Blundell, 2008), while others have not found an association between both variables (e.g. Snoek, Engels, van Strien, & Otten, 2013). Studies measuring actual food intake in response to negative emotions generally show no relation with BMI (e.g. Blissett, Haycraft, & Farrow, 2010; Wallis & Hetherington, 2009). This may be due to the fact that the BMI is determined by numerous factors next to energy intake, such as physical activity and the metabolism of the individual (Sharma & Padwal, 2010). Furthermore, researchers suggest that emotional eating may be associated with other individual characteristics which diminish the risk of obesity, like attempting to restrict eating behaviour (Gibson, 2012; Greeno & Wing, 1994).

Although emotional eating has mostly been studied in adults, research indicates that emotional eating, self-reported as well as actual emotional eating behaviour, is already present in children and adolescents (e.g. Blissett et al., 2010; Braet, Beyers, Goossens, Verbeke, & Moens, 2012; Carper, Fisher, & Birch, 2000; Nguyen-Michel, Unger, & Spruijt-Metz, 2007). In addition, adolescence is a typical period of significant stress and also a crucial time for the development of weight gain (Levitin & Davis, 2010). As such, adolescence seems to be an important period for the prevention and intervention of emotional eating. Therefore researchers have investigated which antecedents may have an impact on this eating style in children and adolescents.

Firstly, affect regulation models postulate that experiencing emotional distress will increase the likelihood of emotional eating, especially in individuals demonstrating deficits in emotion regulation (ER) (Grilo & Shiffman, 1994). ER refers to understanding and managing emotions, which includes the use of strategies for dealing with uncomfortable emotions (Harrist, Hubbs-Tait, Topham, Shriver, & Page, 2013). Survey studies indeed show associations between children’s poor ER on the one hand and emotional eating and binge eating on the other hand (Czaja, Rief, & Hilbert, 2009; Harrist et al., 2013). Secondly, a recent survey study of Vandewalle, Moens, and Braet (2014) showed that parental rejection is associated with emotional eating of obese youngsters and that this relation is mediated by the maladaptive ER of the youngsters. The impact of parental rejection on shaping children’s psychological adjustment had already been suggested by Rohner (1975) in his interpersonal acceptance-rejection theory (IPARTheory;

formerly known as PARTheory). According to the IPARTheory, children have developed a biologically based emotional need to be comforted, supported and cared for by attachment figures. If this emotional need is not met, in other words, if children experience their relationship with attachment figures as rejecting, then they will show overwhelming emotions and receive little support from attachment figures to regulate them, leading to more ER deficits (Meesters & Muris, 2004). This in turn may affect the individuals' eating style, as emotional eating is defined as a form of coping with emotional distress (van Strien, Frijters, Bergers, & Defares, 1986). Vandewalle et al. (2014) specifically found that only maternal rejection and not paternal rejection has an influence on the ER and in turn on the emotional eating of the obese youngster. The authors hypothesised that this may be because both feeding and ER socialisation are, especially in Western societies, an integral part of the mother–child relationship (Kliewer, Fearnow, & Miller, 1996; Winnicott, 1957).

Aim of the present study

In the present study, we wanted to build on the former work of Vandewalle et al. (2014). In this study, a clinical sample of obese youngsters was included, which is a select subgroup with increased psychiatric comorbidity, ER difficulties and disturbed eating (Messerli-Burgy, Pjanic, Jaggi, Fresa, & Znoj, 2012; Zametkin, Zoon, Klein, & Munson, 2004). As research has shown that self-reported emotional eating as well as actual emotional eating behaviour is also present in children and adolescent from the community (Blissett et al., 2010; Vannucci et al., 2013), we wanted to investigate if the relation between parental rejection, maladaptive ER and emotional eating is also present in a community sample of youngsters. Therefore we conducted a longitudinal survey study with two waves, using latent change models (LCM). These models make it possible to look at both the level (at the first wave) and the change (between the two waves) of the variables. We hypothesise that: (1) regarding the level of the variables, higher levels of maternal rejection, and not paternal rejection, will be related to higher levels of maladaptive ER and higher levels of emotional eating of the youngster. Higher levels of maladaptive ER will be related to higher levels of emotional eating. (2) The relation between the level of maternal rejection and the level of emotional eating will be mediated by the level of maladaptive ER. (3) Regarding the change in the variables, changes in maternal rejection, and not paternal rejection, will be positively related to changes in maladaptive ER and changes in emotional eating of the youngster over time. Changes in maladaptive ER will be positively related to changes in emotional eating. (4) The relation between the change in maternal rejection and the change in emotional eating will be mediated by the change in maladaptive ER.

Method

Participants and procedure

The sample comprised 81 youngsters (44.4% boys and 55.6% girls) between the ages of 10 and 16 years ($M = 12.86$ years, $SD = 1.65$), who participated in a larger longitudinal study investigating the relation between parental behaviour and youngsters' eating behaviour. Only mothers with at least two children between the ages of 10 and 16 years old were eligible to participate in this larger study. The participation of both mother and

two children meeting the age range (10–16 years) was an inclusion criteria in the larger study. Concerning the present study, we randomly selected one youngster out of every participating family to avoid correlation between outcomes of siblings. Participants were recruited via two procedures. Twenty-nine youngsters (48.3% boys and 51.7% girls) were recruited via schools. Approximately 1300 flyers were spread in schools with different educational levels, containing information about the study and containing informed consents for both the youngster and their parents. In total 459 families responded to the flyer, of which 68 families consented to participate. Families who declined participation were mostly families not meeting the inclusion criteria (two children in the age range of 10–16 years). After receiving informed consent from parent and youngster, participants were asked via email to fill in an online survey at home. Of the 68 families 39 families were excluded from the study due to non-response at the first wave. Four months after the first email, participants were once again contacted via email to fill in the online survey a second time ($M = 112$ days between study moments). Of the 29 remaining families, 17 families participated a second time (dropout 41.38%).

Due to the high non-response at the first wave, we decided to enlarge the sample, using a different procedure. Concerning this second procedure, 52 youngsters (42.3% boys and 57.7% girls) were recruited by trained psychology students and were asked to fill in questionnaires at home during two home visits. During this home visits, the trained psychology students obtained informed consent from both the youngster and the parent and were at the youngsters' disposal in case of questions regarding the questionnaires. A second home visit took place after one month ($M = 30$ days between study moments). Of the 52 families, only 2 families did not participate a second time (dropout 3.85%).

There were no significant baseline differences between participants recruited by one or the other procedure (controlled for gender, age, adjusted BMI, maternal and paternal rejection, maladaptive ER and emotional eating). There were however baseline differences between participants who did or did not participate a second time, with both baseline maladaptive ER ($t(78) = -2.34, p = .02$) and emotional eating ($t(79) = -2.06, p = .04$) being significantly higher in the participating group. Other variables measured at baseline (gender, age, adjusted BMI and maternal and paternal rejection) were not different between both groups. The study was approved by the Institutional Ethical Committee.

Measures

Anthropometric and demographic measures

The youngsters recruited via the first procedure were asked at baseline to report their weight and length online. The youngsters recruited via the second procedure were measured and weighed by the trained psychology student during the first home visit. The BMI was calculated as weight (kg)/height (m)². We used the adjusted BMI ((actual BMI/percentile 50 of BMI for age and gender) $\times$ 100). This method allows us to compare the BMI of children of different ages and gender. Adjusted BMI below 85% is classified as underweight, above 120% is classified as overweight and above 140% is classified as obese (Van Winckel & van Mil, 2001). The 50th percentiles of the BMI for age and gender are based on normative data in a Flemish sample (Roelants &

Hauspie, 2004). The participants had a mean adjusted BMI of 101.98% (SD = 16.09) with a range of 71.92–154.95% (9.88% missing). To examine possible relations between the adjusted BMI and the study variables, Pearson correlations were computed, however none of the relations were significant.

The familial socio-economic situation was calculated using the Hollingshead Index of Social Position (ISP). The mother of the youngster was asked to fill in the education and occupation of herself and her partner at baseline, which was then used to classify the families in one of five social position indexes, taking the domestic situation of the youngster into account (Hollingshead, 1975). We recoded the five social position indexes into three social classes (high = upper and upper middle; middle; low = lower middle and lower). The lower class holds 10.7% of the families, the middle class 57.3% of the families and the upper class 32% (percentages are valid, 7.4% of mothers did not provide this information). To check for possible relations between the familial ISP-score and the study variables, Pearson correlations were computed. There were no significant correlations.

Maternal and paternal rejection

Maternal and paternal rejection were assessed by means of the EMBU-A (Egna Minnen Beträffande Uppfostran: My memories of child upbringing – Adolescent version; Gerlsma, Arrindell, Vanderveen, & Emmelkamp, 1991), a Dutch adolescent version of the original EMBU (Perris, Jacobsson, Lindstrom, Knorrning, & Perris, 1980). The original EMBU is a self-report measure, intended to assess adults' recollections of their parents' child-rearing behaviour. In designing the EMBU-A, the items of the original adult EMBU were adapted for use with young adolescents. This adaptation creates the opportunity to obtain current perceptions of parenting while living at home. The EMBU-A consists of 56 items assessing four domains of parental behaviour: emotional warmth, rejection, overprotection and favouring subject. To limit the load on the participants only the subscale rejection (19 items concerning mother and 19 items concerning father, e.g. 'Is your mother/father ever harsh and unfriendly to you?') was assessed in the present study. All of the items are behaviour-oriented in their formulation, as such excluding an attributional evaluation. Items were to be answered for mother and father separately, on four-point Likert scale from 1 = No, never to 4 = Yes, most of the time. Analyses of Gerlsma et al. (1991) showed a good internal consistency reliability and construct validity for the subscale rejection. In this study, the Cronbach α for maternal rejection and paternal rejection were .92 and .94 at time 1 (T1) and .91 and .94 at time 2 (T2), respectively.

Emotion regulation of the child

The ER of the child was assessed by means of the FEEL-KJ (Fragenbogen zur Erhebung der Emotionregulation: bei Kindern und Jugendlichen; Questionnaire to Assess Children's and Adolescents' ER Strategies; Braet, Cracco, & Theuwis, 2013; Grob & Smolenski, 2005). The FEEL-KJ is a 90-item self-report measure used to assess 15 ER strategies in response to three emotions; anxiety, sadness and anger. In this study, only ER strategies in response to sadness and anger were included in order to limit the load on the participants. The emotions sadness and anger were chosen

considering that sadness has been numerous associated with emotional eating (e.g. Goldschmidt, Tanofsky-Kraff, & Wilfley, 2011; Macht & Mueller, 2007) and that hostility and aggression have been associated with paternal rejection (Rohner, 2004). The FEEL-KJ measures primary strategies and two secondary scales revealed by factor analysis: an adaptive strategies scale (seven strategies: problem-oriented action, cognitive problem-solving, acceptance, neglect, distraction, revaluation and put into good humor) and a maladaptive strategies scale (five strategies: giving up, withdrawal, aggressive action, self-devaluation and perseveration). Three additional strategies (social support, expression and control over emotion) are not part of the secondary scales, but considered separately. The maladaptive scale, entailing the emotions sadness and anger (20 items; e.g. ‘When I’m sad/angry, I don’t want to see anyone’), was used in this study. Items were to be answered on five-point Likert scale from 1 = Almost never to 5 = Almost always. Analyses showed good internal consistency, with Cronbach α between .69 and .91 for the ER strategies. In this study, the Cronbach α for the maladaptive ER scale was .87 at time 1 (T1) and .88 at time 2 (T2).

Emotional eating of the child

The Dutch Eating Behaviour Questionnaire – child version (DEBQ; Braet et al., 2008; van Strien et al., 1986) was used to assess the emotional eating of the youngsters. The DEBQ contains 33 items, assessing the presence of three types of disturbed eating behaviour: restrained eating, external eating and emotional eating. Only the subscale emotional eating (13 items; e. g. ‘Do you have a desire to eat when you are emotionally upset?’) was assessed in the current study. Items are formulated as specific eating behaviours and are to be rated on their frequency of occurrence on a five-point Likert scale from 1 = never to 5 = very often. Studies have indicated the usefulness of the Dutch version of the DEBQ in children and adolescents between the age of 7 and 17 years. Research showed a stable factor structure, satisfying internal consistency reliability and good test–retest reliability (Braet et al., 2008). Further, research showed a good external validity for the DEBQ in children (Ricciardelli & McCabe, 2001). In the present study, the coefficient α for the subscale emotional eating was .94 at both time moments.

Analytic plan

Missing data analysis was conducted as there was dropout during the study (6.8% missing data points). The Little (1988) MCAR test was not significant [$\chi^2(326) = 162.40$, $p = 1.00$], indicating that the missing data can be considered as being completely at random. The missing values were estimated using maximum likelihood estimation. First, descriptive statistics were calculated for the study variables measured at baseline. Next, for each construct three-item parcels were created using Confirmatory Factor Analysis. Multiple indicators of each construct are necessary to reliably estimate level and change of each construct.

Subsequently, level and change of maternal and paternal rejection, maladaptive ER strategies and emotional eating were estimated using LCMs (see Figure 1 for example; model of maternal rejection) (LCMs; Hertzog & Nesselroade, 2003; McArdle & Nesselroade, 1994). The estimated change of the constructs is the absolute change over the two measurements in the latent variables. In this way, the problem of measurement

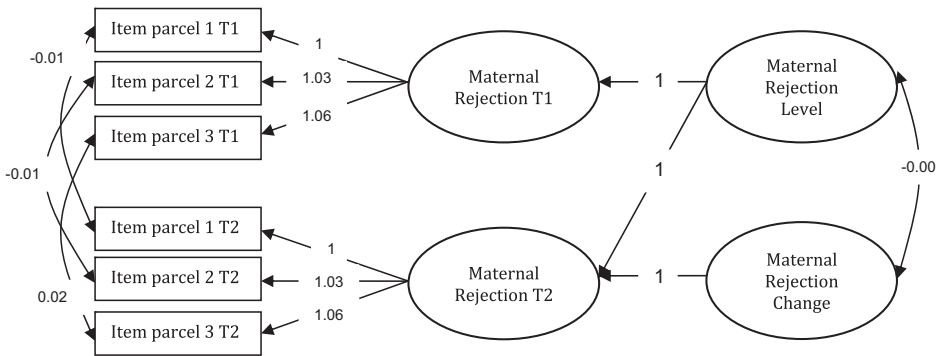


Figure 1. Unstandardised LCM of maternal rejection measured at Waves 1 and 2.

error with simple difference scores is avoided (Rogosa, Brandt, & Zimowski, 1982). In the case of maternal rejection, for instance (see Figure 1), corresponding factor loadings of the three measures of maternal rejection are set equal across time and the corresponding residual errors of the three measures are allowed to co-vary across time, which reflects stability in systematic errors over time. The fixed 1 regression coefficients involving the latent variables implicitly define the maternal rejection Level latent variable as equal to maternal rejection at Wave 1 (maternal rejection T1) and the maternal rejection Change latent variable as the difference between maternal rejection at the two waves (maternal rejection T2- maternal rejection T1). The LCM also allows to estimate the mean and variance of the latent level factor and change factor. These variances indicate interindividual differences in level and change. In that way, it is possible to examine how the Level and Change latent variables of the constructs are related via Pearson's product moment correlations. Furthermore, bootstrapping was used in structural equation modeling (SEM) to analyse the mediation effects (Muthén & Muthén, 1998–2012). The 95% confidence intervals around the estimates were examined. Confidence intervals not including zero indicate significant indirect effects (MacKinnon, Lockwood, & Williams, 2004). Data were analysed using the SPSS version 22 and Mplus version 7.3 (Muthén & Muthén, 1998–2012). *P*-values < .05 were considered statistically significant.

Results

Descriptive statistics for study variables

Descriptive statistics were calculated for the study variables measured at baseline. Regarding maternal and paternal rejection, the mean scores were 1.38 (SD = .40) and 1.42 (SD = .51), respectively. These scores are comparable with unpublished normative data of 1153 Flemish youngsters ($t(80) = .39, p = .70$; $t(80) = 1.18, p = .24$). Respectively 9.9 and 16% of our sample reported to never have experienced maternal or paternal rejection (mean score = 1). In addition, 17.3 and 16% of our sample had a high score ($> M + 1SD$) on the maternal and paternal rejection subscales, in comparison with the normative data.

With regard to the maladaptive ER scale, the mean score was 2.60 ($SD = .61$). All participants in our sample reported using maladaptive ER strategies to some extent (mean score > 1). Concerning emotional eating, the mean score was 1.85 ($SD = .70$) which corresponds with results found in a similar community study ($t(80) = -.87$, $p = .39$) as well as with normative data ($t(80) = -.61$, $p = .54$) (Michels et al., 2012; van Strien & Oosterveld, 2008). In our sample, 14.8% of the youngsters reported to have never experienced emotional eating (mean score = 1), while the same percentage (14.8%) of the youngsters scored above the 80th percentile of the normative data, which classifies them as ‘emotional eaters’ (van Strien & Oosterveld, 2008).

Preliminary analyses

Univariate LCMs were estimated for each construct. Table 1 offers a summary of the fit indices and parameter estimates of the models. For the constructs maternal rejection, paternal rejection and maladaptive ER strategies the LCMs fitted the data adequately. For the construct emotional eating, the model fit was less adequate according to the root mean square error of approximation (RMSEA). When looking at the mean change, paternal rejection showed a significant decrease over time, while maternal rejection did not change. Further, maladaptive ER strategies and emotional eating showed significant decreases over time, averaged across all participants. Next to the mean change, results showed a significant variance for all the constructs regarding to initial levels and for all constructs, except for paternal rejection, regarding the change. This indicates significant interindividual differences in initial level and change over time for these variables. Concerning the variance of change, individual estimates of true change for all variables ranged from negative values (declines) to positive values (increases).

Correlations among study variables

A Pearson correlation matrix was constructed containing the level and change latent variables of the constructs (see Table 2). Regarding the level of the variables, results showed that the level of maternal rejection was positively related to the level of maladaptive ER strategies and emotional eating. In addition, the level of maladaptive ER strategies was positively related to the level of emotional eating. The level of paternal rejection was not related to the level of maladaptive ER strategies nor to the level of emotional eating.

Concerning the change in the variables, results showed that the change in maternal rejection was neither related to the change in maladaptive ER strategies nor to the change in emotional eating. Like maternal rejection, the change in paternal rejection was not related to the change in maladaptive ER strategies nor to the change in emotional eating. Furthermore, the change in maladaptive ER strategies was not related to the change in emotional eating.

Mediation analysis

Based on the significant Pearson correlations between the study variables, the indirect effect of the level of maternal rejection on the level of emotional eating through the level of maladaptive ER strategies was examined. The model provided a good fit to the

Table 1. Fit indices and parameter estimates of univariate LCMs of constructs.

	Fit indices			Parameter estimates			
	df	SBS- χ^2	RMSEA	CFI	Level		Change
					M	s^2	
Maternal rejection	11	11.85	.03	.99	1.33***	.14**	-.02
Paternal rejection	11	12.81	.05	.99	1.36***	.22***	-.06*
Maladaptive ER strategies	11	11.24	.02	.99	2.62***	.31***	-.15*
Emotional eating	11	33.54***	.16	.93	1.89***	.47***	-.25**
							-.29***
							-1.38-1.29

Note: SBS- χ^2 – Satorra Bentler scaled chi-square; RMSEA – root mean square error of approximation; CFI – comparative fit index. ER – emotion regulation.

* $p < .05$; ** $p < .01$; *** $p < .001$.

Table 2. Pearson correlations between level and change latent variables.

		Maternal rejection		Paternal rejection		Maladaptive ER strategies		Emotional eating	
		Level	Change	Level	Change	Level	Change	Level	Change
Maternal rejection	Level	–							
	Change		.06	.54***	–.64***	.39***	.03	.23*	–.01
Paternal rejection	Level			–.09	.25*	.10	–.01	–.06	–.11
	Change				–.64***	.20	–.03	.01	.07
Maladaptive ER strategies	Level					–.17	–.11	.08	–.06
	Change						.07	.27*	–.12
Emotional eating	Level							–.11	.18
	Change								–.44***
									–

Note: ER = emotion regulation.

* $p < .05$; *** $p < .001$.

data, $\chi^2(1) = 1.56$, $p = .21$, RMSEA = .08, CFI = .97. The indirect effect was marginally significant ($b = .19$, $p = .07$; 95% CI [–.018, .400]).

Discussion

This study aimed to test the relation between parental rejection, maladaptive ER and emotional eating in a community sample of youngsters. Therefore, we conducted a longitudinal study with two waves and estimated level and absolute changes in these variables using LCMs. Via Pearson's product moment correlations the relations between the variables were investigated. In addition, indirect effects were examined via SEM with bootstrapping. When looking at the level of the variables, results show that higher levels of maternal rejection are related to higher levels of maladaptive ER strategies and to higher levels of emotional eating. The level of paternal rejection, however, is not related to the level of maladaptive ER strategies, nor to the level of emotional eating. Furthermore, higher levels of maladaptive ER strategies are related to higher levels of emotional eating. The indirect effect of the level of maternal rejection on the level of emotional eating through the level of maladaptive ER strategies is marginally significant. The results approximate the results of Vandewalle et al. (2014) in a referred sample of obese youngsters. This suggest that also in the general population, youngsters reporting higher maternal rejection use more maladaptive ER strategies and show more emotional eating.

Concerning the change in the variables, results show that the change in maternal rejection is not related to the change in maladaptive ER strategies, nor to the change in emotional eating. This may be due to the lack of change in maternal rejection in our sample. However, even though maternal rejection did not change over time, maladaptive ER strategies and emotional eating did change significantly, which suggest the additional influence of other predictors. The influence of parents may diminish when children grow older and instead the influence of important others may perhaps increase. The emotional socialisation by parents, such as helping the child to learn how to deal with troublesome situations, already starts in the infancy (Kopp, 1989). As children

grow older they rely less on parents and more often rely on other socialisation agents such as peers (Eisenberg & Morris, 2002; Morris, Silk, Steinberg, Myers, & Robinson, 2007; Silk, Steinberg, & Morris, 2003). Research does confirm that peer rejection in adolescence is related to the use of maladaptive ER strategies of the rejected adolescents (Scholte & Van Aken, 2006). In addition, a diary study has shown that fluctuations in daily interpersonal hassles are associated with fluctuations in snacking. Therefore, it might be interesting to investigate the relation between parental and peer rejection and emotional eating in a diary study, looking at day-to-day within-person fluctuations (O'Connor et al., 2008).

Further, the results show that the change in paternal rejection is not related to the change in maladaptive ER strategies nor to the change in emotional eating. This is in line with our hypothesis. However, this might also be explained by the low variance in the change of paternal rejection in our sample.

In addition, the change in maladaptive ER strategies is not related to the change in emotional eating. This may possibly be explained by the fact that not all individuals having difficulty regulating their emotions turn to food as comfort. Other strategies may be used to cope with negative emotions, like aggression or alcohol abuse (Röll, Koglin, & Petermann, 2012; Sher & Grekin, 2007). As such, a change in maladaptive ER strategies in some individuals may not be related to a change in emotional eating, as emotional eating is non-existent in these individuals.

This study has some limitations. Although the study is longitudinal, introducing a third study moment would have been optimal. In this way, it would have been possible to measure each variable at a different study moment and to test a longitudinal mediation model using Latent Growth Curve Models (LGCM; Duncan & Duncan, 2004; McArdle & Nesselroade, 2002; von Soest & Hagtvet, 2011; Willett & Sayer, 1994). Moreover, as parental rejection may possibly have more influence on the ER of young children, it would be interesting to measure parental rejection and ER at a younger age and investigate if this has an enduring effect on emotional eating in the adolescence. In addition, it may be interesting to also look at the other side of the continuum, by including parental warmth next to parental rejection. Results from previous research suggest that parental warmth may enhance the emotional self-regulation capacities of the child (Baker & Hoerger, 2012).

Further, the exclusive reliance on self-report measures in this study may be problematic, especially regarding the assessment of ER and emotional eating of the youngsters. Concerning parental rejection however, Rohner (2004) stresses the importance of the youngster's perception in comparison with parents' perception or objective observations; a child may feel unloved and this in turn may affect the child's mental health, even if outside observers fail to detect parental rejection. But regarding ER and emotional eating, some researchers have argued that the youngster's perceptions may be influenced by the individual's ability to assess their own behaviour (e.g. Evers, de Ridder, & Adriaanse, 2009; Tull, Barrett, McMillan, & Roemer, 2007). Concerning emotional eating, Bongers, Jansen, Houben, and Roefs (2013) state that people are generally rather poor at recalling their emotions, their eating behaviour and the associations between the two. As previously suggested, it therefore may be interesting to investigate the relation between these variables in a diary study, assessing current emotions and actual food intake. Finally, this study suffered from a low response rate and a high dropout rate, mainly due to the strict inclusion criteria of the larger study.

When looking at current childhood obesity prevention programmes, the focus on parents is often limited to educating parents about healthy eating patterns and reducing sedentary lifestyle of their children (Flodmark, Lissau, Moreno, Pietrobelli, & Widhalm, 2004). Considering the link between the level of maternal rejection and the level of emotional eating, prevention and intervention programmes might focus on assessing and improving the emotional bond between parent and youngster. Furthermore, prevention and intervention programmes might also focus on improving ER considering the link between the level of maladaptive ER strategies and the level of emotional eating. Training in ER was already found to be effective in improving coping and reducing binge eating amongst women with BED (Clyne & Blampied, 2004). Furthermore, Groves, Backer, van den Bosch, and Miller (2012) conducted a review on the effectiveness of dialectical behaviour therapy (DBT) for adolescents, of which ER training is a substantial part. They reviewed the literature on the use of DBT for adolescents with various psychological disorders and behaviours. They conclude that the overall results seem promising, but also note that there is need for further randomised controlled trials. Studies on the effectiveness of ER training in adolescents with an emotional eating style is lacking.

Disclosure statement

The authors declare no conflict of interest.

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